

Setting Up Python and VS Code for Streamlit

For the Streamlit workshop, we will use:

- **Visual Studio Code** to write, edit, and run Python code.
- **Python** to build Streamlit applications.
- **GitHub** to store and share project files and support the deployment of Streamlit applications. If you do not already have a GitHub account, you can [create one for free](#).

This guide will walk you through installing and setting up the tools needed for the workshop. Specifically, you will:

- Install Visual Studio Code
- Install Python 3
- Install the Python extension for Visual Studio Code
- Create a Python virtual environment
- Install Streamlit and other required Python packages

macOS users will also install **Homebrew**, which will be used to install Python.

Resources

- [Getting Started with Python in VS Code](#)
 - [Installing Visual Studio Code on macOS](#)
 - [Installing Visual Studio Code on Windows](#)
 - [Installing Streamlit Using the Command Line](#)
 - [Python Virtual Environments](#)
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1. Install Visual Studio Code

Go to the [Visual Studio Code download page](#).

For macOS Users

1. Download the macOS version.
2. Open the downloaded `.dmg` file.
3. Drag **Visual Studio Code** into the **Applications** folder.

4. Open Visual Studio Code from the Applications folder.

For Windows Users

1. Download the Windows version.
 2. Open the downloaded `.exe` installer.
 3. Follow the installation prompts.
 4. When the options appear, it is helpful to select **Add to PATH** and **Add “Open with Code” action**.
 5. After the installation is complete, open Visual Studio Code from the Start menu.
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2. Install Python 3

For macOS Users

On macOS, we will use **Homebrew** to install Python. Homebrew is a package manager that makes it easier to install software on a Mac.

2.1. Install Homebrew

1. Open Terminal.

You can find Terminal by pressing **Command + Space**, searching for **Terminal**, and pressing Enter.

2. Copy and paste the following command into Terminal:

```
/bin/bash -c "$(curl -fsSL  
https://raw.githubusercontent.com/Homebrew/install/HEAD/install.sh)"
```

3. Press Enter and follow the instructions displayed in Terminal.

You may be asked to enter your Mac password. When you type your password, no characters will appear on the screen. This is normal. Type your password and press Enter.

4. After the installation is complete, carefully follow the **Next steps** displayed in Terminal.

The commands may look similar to the following:

Do not copy the commands below. They are just examples. Use the commands displayed on your own computer.

```
echo 'eval "$(/opt/homebrew/bin/brew shellenv)"' >> ~/.zprofile  
eval "$(/opt/homebrew/bin/brew shellenv)"
```

Do not copy the commands above. They are just examples. Use the commands displayed on your own computer.

5. Confirm that Homebrew is working:

```
brew --version
```

You should see output beginning with something similar to:

```
Homebrew 6.x.x
```

The exact version number may be different.

2.2. Install Python

In Terminal, run:

```
brew install python
```

Confirm that Python was installed:

```
python3 --version
```

You should see output similar to:

```
Python 3.x.x
```

The exact version number may be different.

For Windows Users

Because I do not have access to a Windows computer to test the installation process, please follow the Windows instructions in [Getting Started with Python in VS Code](#) guide.

3. Install the Python Extension for Visual Studio Code

The Python extension provides Python language support and allows Visual Studio Code to detect and switch between Python environments.

1. Open Visual Studio Code.
2. Select the **Extensions** icon in the left sidebar.
3. Search for **Python**.
4. Install the extension named **Python**, published by **Microsoft**.



4. Create a Project Folder

First, create a folder for your Streamlit project. For example, you might create a folder named:

```
Streamlit
```

The folder can be located anywhere on your computer, but make sure you remember where you saved it.

Next, open Visual Studio Code and select:

```
File → Open Folder
```

Select the folder you created, and then choose **Open**.

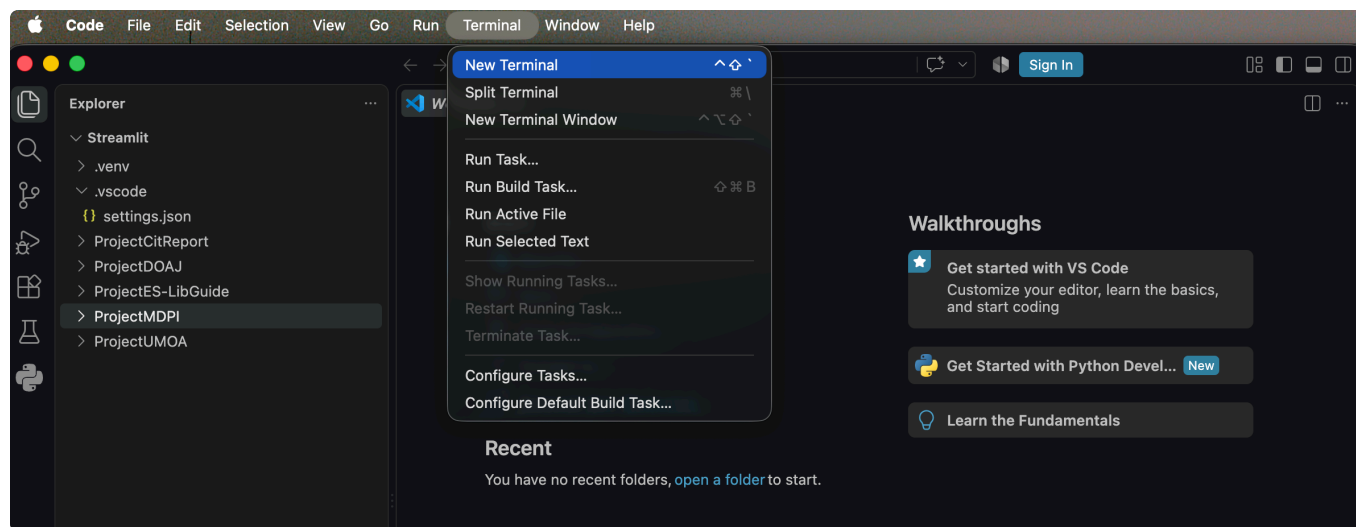
It is important to open the entire folder rather than an individual Python file. Visual Studio Code will use this folder as your project workspace and will store your virtual environment, Python files, and other project materials inside it.

5. Create a Virtual Environment

It is recommended to create a **virtual environment** for your Streamlit project. A virtual environment keeps the packages used by one project separate from packages used by other projects.

Open the Terminal in your Visual Studio Code. To open the Terminal, in Visual Studio Code, select:

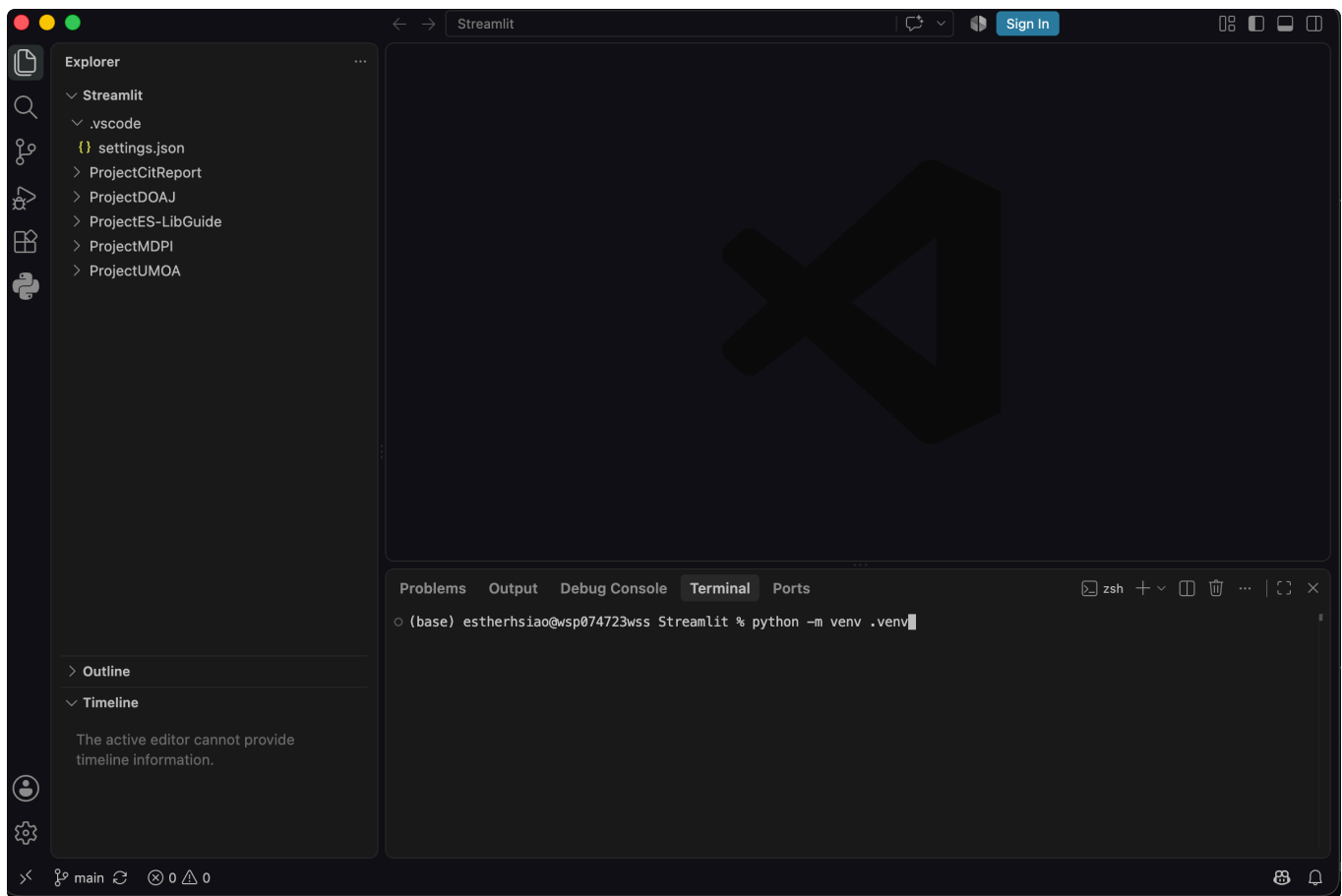
```
Terminal → New Terminal
```



The terminal should open at the bottom of the window.

Then run the command below to create a virtual environment named `.venv` inside the project folder.

```
python -m venv .venv
```



Your folder will now contain:

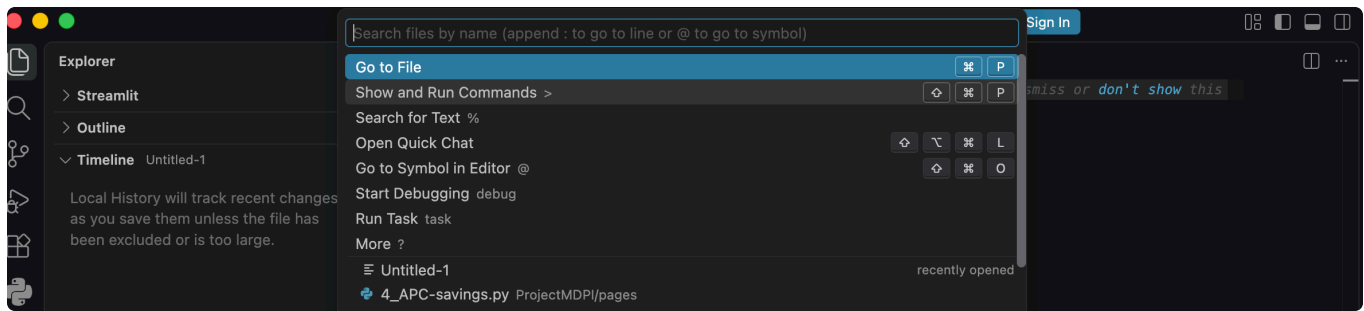
```
streamlit/  
└── .venv/
```

The `.venv` folder contains a separate Python interpreter and a location for the packages used by this project.

6. Select the Virtual Environment in Visual Studio Code

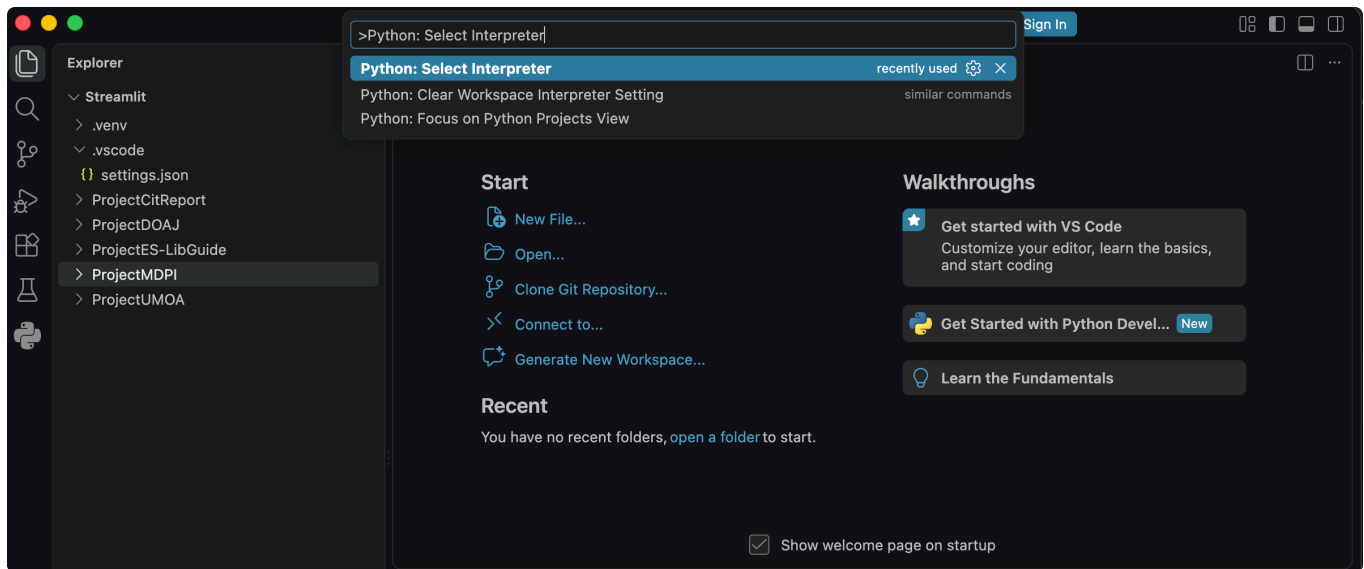
In the search bar at the top of Visual Studio Code, select **Show and Run Commands**, or use the keyboard shortcut:

- **macOS:** Command + Shift + P
- **Windows:** Ctrl + Shift + P

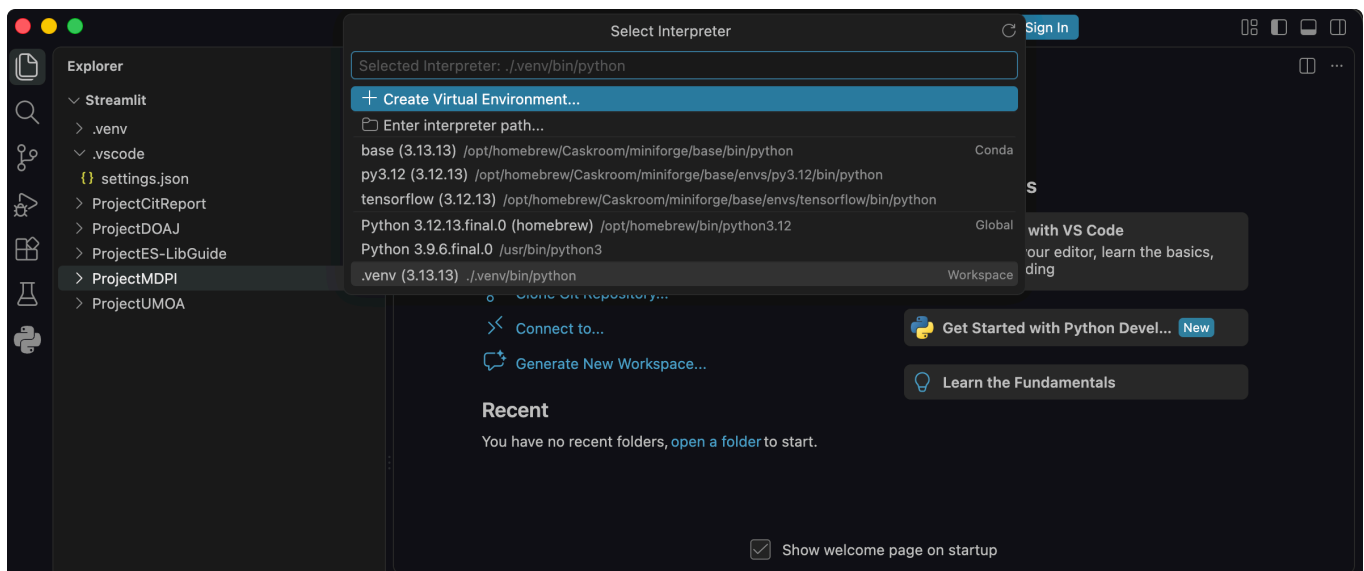


Search for:

Python: Select Interpreter



Select the Python interpreter associated with the `.venv` virtual environment. It may appear as `.venv (3.13.x)`.



7. Restart Terminal in Visual Studio Code

In Visual Studio Code, select:

```
Terminal → New Terminal
```

The terminal should open at the bottom of the window.

If Visual Studio Code selected the virtual environment correctly, the prompt should begin with:

```
(.venv)
```

For example, my prompt is like:

```
(.venv) (base) estherhsiao@wsp074723wss Streamlit %
```

If it does not, activate the environment manually:

```
source .venv/bin/activate
```

11. Install Python Packages

Only install the packages after the virtual environment has been activated.

First, update pip:

```
python -m pip install --upgrade pip
```

Then install the packages used in the workshop:

```
python -m pip install streamlit numpy pandas matplotlib plotly seaborn
```

Confirm that Streamlit was installed:

```
python -m streamlit --version
```

You can also view all installed packages:


```
python -m pip list
```

12. Test the Streamlit Installation

Run Streamlit's example application:

```
python -m streamlit hello
```

Streamlit should start a local server and open the example application in your web browser. Streamlit recommends this command as a way to test the installation.

To stop the Streamlit server, return to Terminal and press:

```
Control + C
```